

Ryan Qin

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Education

University of Waterloo

Sept 2024 - Jun 2029

Bachelor of Applied Science in **Electrical Engineering**, Honours, Co-op

Technical Skills

Languages: C, C++, C#, Assembly, Verilog, Java, Python, HTML/CSS, JavaScript

Technologies: React, Node.js, MongoDB, TCP/IP, Linux, Robotics, STM32, DSP

Development Tools: Git, Android, AWS, Fusion 360, Altium, Cadence, TensorFlow, MATLAB, DSO

Professional Experiences

I/O Compliance Engineer | TensorFlow, OpenCL, C++, C#, Cadence, DSO, *I²C*

May 2026 - Present

AMD

- Developing a **TensorFlow** model for predicting root cause of **DP** and **HDMI** test failure, aiming to reduce debugging time by **50%**
- Applying **autoencoder ML** to HDMI protocol analyzer for **improving** effectiveness of **detecting** frame glitches and stutters compared to **standard** test patterns and manual monitor observations
- Designing **STM32-based I²C** module for **automatic** HDMI **EDID** setting, helping to reduce manual rewiring work
- Optimizing electrical Compliance analysis calculations using **OpenCL**, utilizing GPU parallelism for consecutive test runs

Electrical Engineer | Altium, LTspice, STM32, C++, TCP/IP

Sep 2025 - Dec 2025

ATS Corporation

- Managed **full-cycle** design and verification for **four parallel** electronics projects, ensuring **on-time, spec-compliant** delivery
- Designed and tested an inductively isolated pulse conditioner board in Altium and LTspice, protecting PLC against stray voltage while reducing 80% of Sub-GHz noise from turbocharger's hall-effect tachometer
- Designing robotic arm assembly line's **E-Stop** interlock and area scanner systems, ensuring a **near-zero** residual risk
- Configuring jet nozzle spray-pattern analyzer cameras using **OpenCV**, aiming to increase test-efficiency by 60%

Electrical Coordinator | C#, .Net, Visual Studio, MicroStation, Python

Jan 2025 - May 2025

WSP

- Recognized by **WSP Canada's** Rail and Transit VP for OCS engineering Software's **innovation intake** potential
- Independently developed Overhead Contact System (OCS) engineering software in **C#**, reducing Calculation time by **40%**
- Optimized traction power substation's cooling system layout in **COMSOL FEA**, aimed to extend transformer life by 10 years
- Implemented AI-driven weather prediction using **Prophet**, improving storm and temperature tolerance by **30%**

Projects

FPGA Clock 📄 | Verilog, Vivado

Dec 2025 - Jan 2026

- Developed a **real-time** VGA digital clock for FPGA using **Verilog**, designing the logic for time counters, VGA signaling, and character rendering using **Vivado** and **HDL simulation**
- Implemented parameterized **RTL** to simplify character resizing and repositioning across display resolutions

Autodump 📄 | I/O system Design, C++, TCP/IP, SSH, Python, OpenCV

Nov 2025

- Developed an autonomous self-emptying trash robot, achieving an overall 3rd place at MakeCU
- Engineered an Arduino-based serial distributed I/O system, improving software-hardware integration efficiency while maintaining pulse signal integrity for servo and ultrasonic sensors
- Designed **ArUco** marker navigation system in **OpenCV**, allowing the robot to **autonomously** navigate back to home position

Competition RC 📄 | Power Electronic Design, Altium Designer, Assembly

May 2025

- Built an RC car with 50% recycled material; selected as the top qualified finalist out of 15 teams
- Designed a custom airsoft motor controller using MOSFET stages, relays, and Arduino for PWM signal conversion, achieving a fast lap record of 17 seconds compared to average 25 seconds lap time
- Engineered safe state logic on Arduino's assembly level firmware, reducing 70% of unexpected startup behavior risks

Baymax Bot 📄 | Machine Learning, TCP/IP, Linux, Python, C++, UART, FPGA

May 2025

- Trained a voice model using Applio, replicating BayMax's Voice with customized response from ChatGPT's API
- Developed an **FPGA** safety controller integrated with Raspberry-Pi TCP/IP system for AI-driven A/V and motor control